Lecture 27: Markov Chains III

- It turns out that many types of Markov chain have rather consistent long-term behaviour.
- In this lecture we shall examine this issue in detail.
- This will lead us to consider classify the states of a Markov chain to reflect the long run properties of the process.
- For the Markov chains with this consistent long-term behaviour (described by an equilibrium distribution), we will characterise the **stationary distribution**: a probability vector that doesn't change with time.
- Equilibrium and stationary distributions are the same for regular Markov chains,

It All Looks the Same from a Distance...

- Recall the Markov chain model for Palmerston North summer rainfall.
- In judging whether it will on day $t = 1$, it is important to know the weather on day $t = 0$.
  - If day $t = 0$ is dry then rain is fairly unlikely on day $t = 1$ (probability 0.13).
  - If day $t = 1$ is rainy then its a 50:50 chance that day $t = 1$ will also be wet.
- Similarly, for assessing whether it will rain on day $t = 2$, initial weather is important.
- But now consider judging whether it will rain on day $t = 30$, or day $t = 1000$ (if the model applied to an endless summer...).
- We would expect the rainfall status on day $t = 0$ to have almost no bearing on the weather so far in the future.

Example: Long Term Weather Forecasting!

- We use our standard Markov chain model for Palmerston North summer rainfall to compute probability of rain on days $t = 1, 5, 50$ and 1000 (note this is different to simulations!)
- Consider two initial conditions:
  - (i) day 0 is dry, so $\pi^{(0)} = (1, 0)$.
  - (ii) day 0 is rainy, , so $\pi^{(0)} = (0, 1)$.
- R snippet below creates initial distributions and transition matrices.

R Snippet

```
> library(expm)
> pi0.dry <- t(c(1,0))
> pi0.wet <- t(c(0,1))
> P <- matrix(c(0.87,0.5,0.13,0.5),nrow=2)
> P5 <- P%^%5
> P50 <- P%^%50
> P1000 <- P%^%1000
```

Long Term Weather Forecasting example continued

R Snippet

```
> pi1.dry <- pi0.dry%*%P
> pi1.wet <- pi0.wet%*%P
> pi1.dry
     [,1] [,2]
[1,] 0.87 0.13
> pi1.wet
     [,1] [,2]
[1,]  0.5  0.5
> pi5.dry <- pi0.dry%*%P5
> pi5.wet <- pi0.wet%*%P5
> pi5.dry
          [,1]      [,2]
[1,] 0.7950817 0.2049183
> pi5.wet
          [,1]      [,2]
[1,] 0.7881473 0.2118527
```

Long Term Weather Forecasting example continued

R Snippet

```
> pi50.dry <- pi0.dry%*%P50
> pi50.wet <- pi0.wet%*%P50
> pi50.dry
          [,1]      [,2]
[1,] 0.7936508 0.2063492
> pi50.wet
          [,1]      [,2]
[1,] 0.7936508 0.2063492
> pi1000.dry <- pi0.dry%*%P1000
> pi1000.wet <- pi0.wet%*%P1000
> pi1000.dry
          [,1]      [,2]
[1,] 0.7936508 0.2063492
> pi1000.wet
          [,1]      [,2]
[1,] 0.7936508 0.2063492
```

Long Term Weather Forecasting example continued

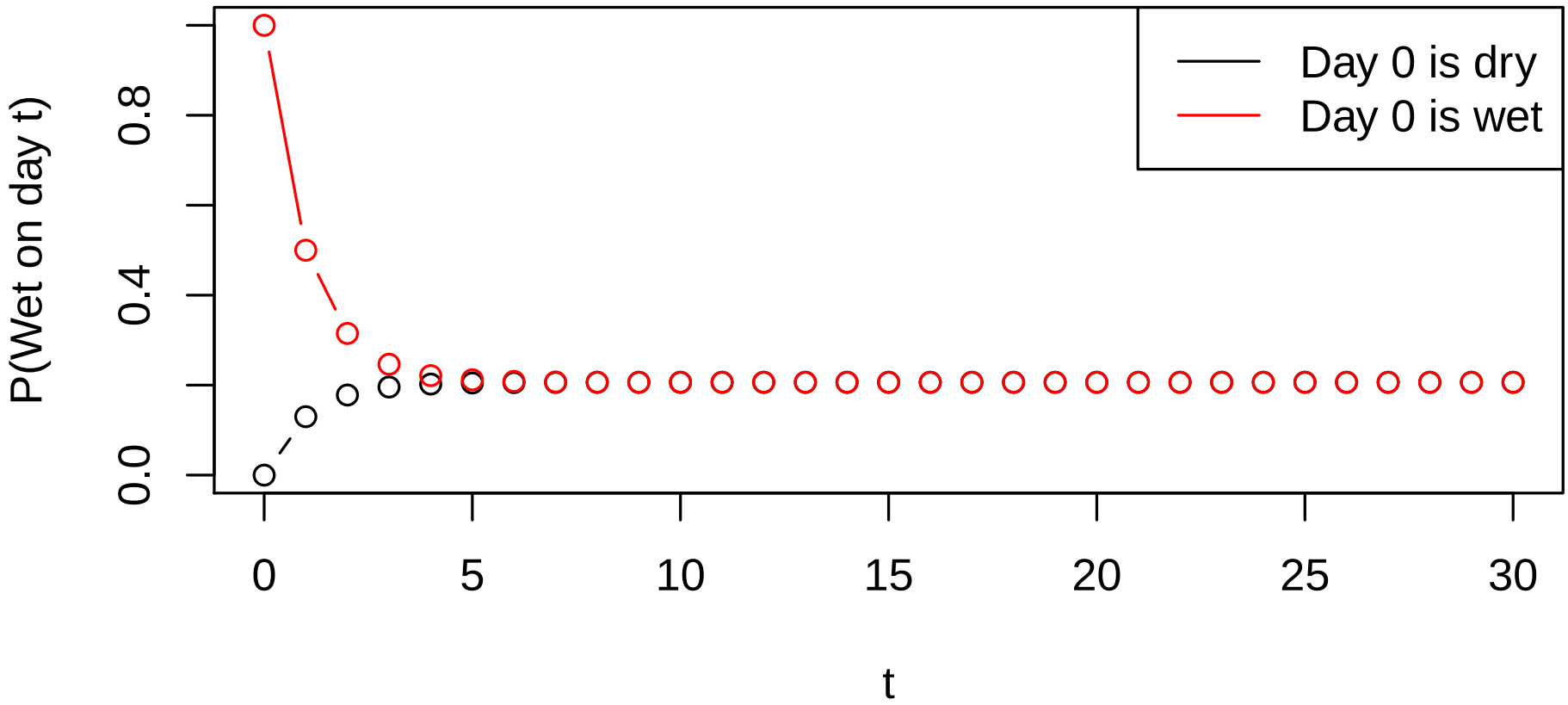
Points to Note

- Even by day $t = 5$, probability of a rainy day is quite similar under the two different initial conditions.
  - If dry on day 0 $\mathrm{P}(Y_5 = 1 | Y_0 = 0) = 0.205$.
  - If wet on day 0 $\mathrm{P}(Y_5 = 1 | Y_1 = 1 = 0.212$
- By day 50 (and same for $t = 1000$) probabilities of rain under different initial conditions are identical to 7 decimal places:

$$\mathrm{P}(Y_{50} = 1 | Y_0 = 0) \approx \mathrm{P}(Y_{50} = 1 | Y_0 = 1) = 0.2063492.$$

Long Term Weather Forecasting example continued

Plot of probability of rain for different initial conditions



Equilibrium Distributions

- We have seem that the initial conditions are eventually forgotten in this Example.
- Moreover, the probability distribution $\pi^{(t)}$ settles down as t becomes large.
- The limiting case of $\pi^{(t)}$ as $t \to \infty$ is called the **equilibrium distribution**.
- This limiting distribution also turns out to be the **stationary distribution**, for reasons we shall see later.
- Many Markov chains share the same kind of consistent long term behaviour as the rainfall model, and hence converge to a equilibrium distribution.
- However, certain conditions on the Markov chain are required to ensure this happens.

Classification of States

- One of the reasons why we can expect a Markov chain like the rainfall model to have consistent long term behaviour is that we can be confident that both states will keep on recurring.
  - It may be wet today, but we know a dry day will follow at some point.
- However, such behaviour is not assured for all systems we might wish to model.
- Specifically, we may find that some states are never visited again after some time point.

Example: The gamblers

- Two gamblers playing a sequence of games of chance.
- Each gambler starts with \$2, and wins or loses \$1 on each game.
  - Games are fair, so probability winning is $\frac{1}{2}$.
- State space $S = \{0, 1, 2, 3, 4\}$ can be fortune Y_t of first gambler.
- $\{Y_t : t = 0, 1, 2, \ldots\}$ is a random walk until boundary at 0 or 4 hit (when one gambler goes broke).



Gamblers example continued

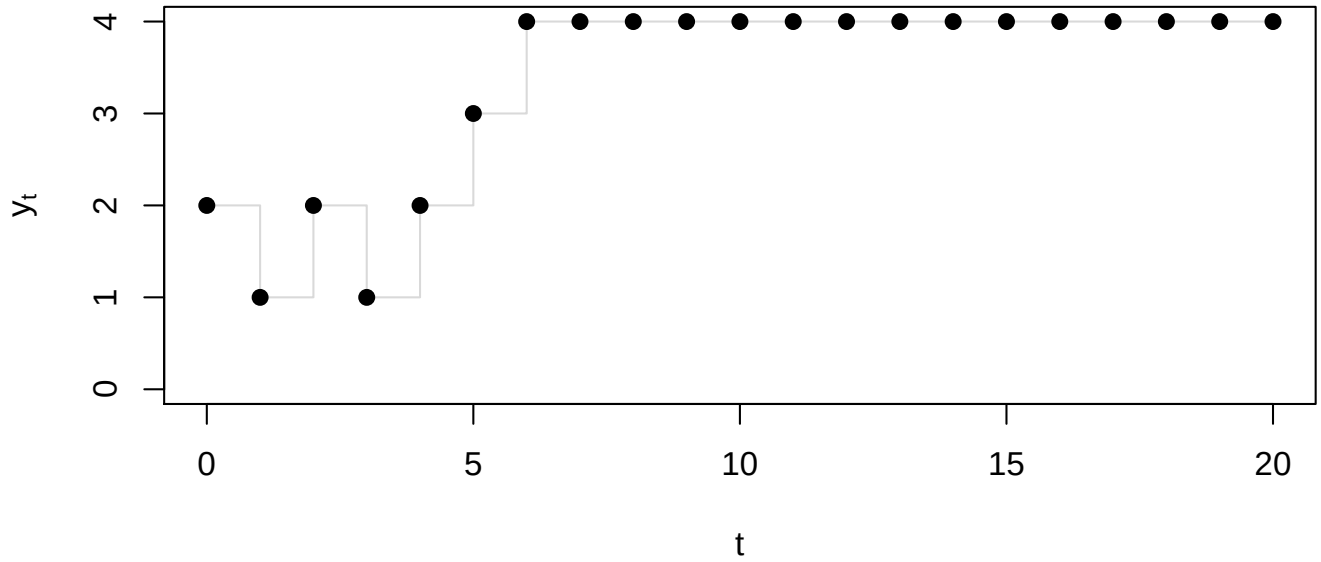
- $\{Y_t : t = 0, 1, 2, \ldots\}$ is a Markov chain with transition matrix

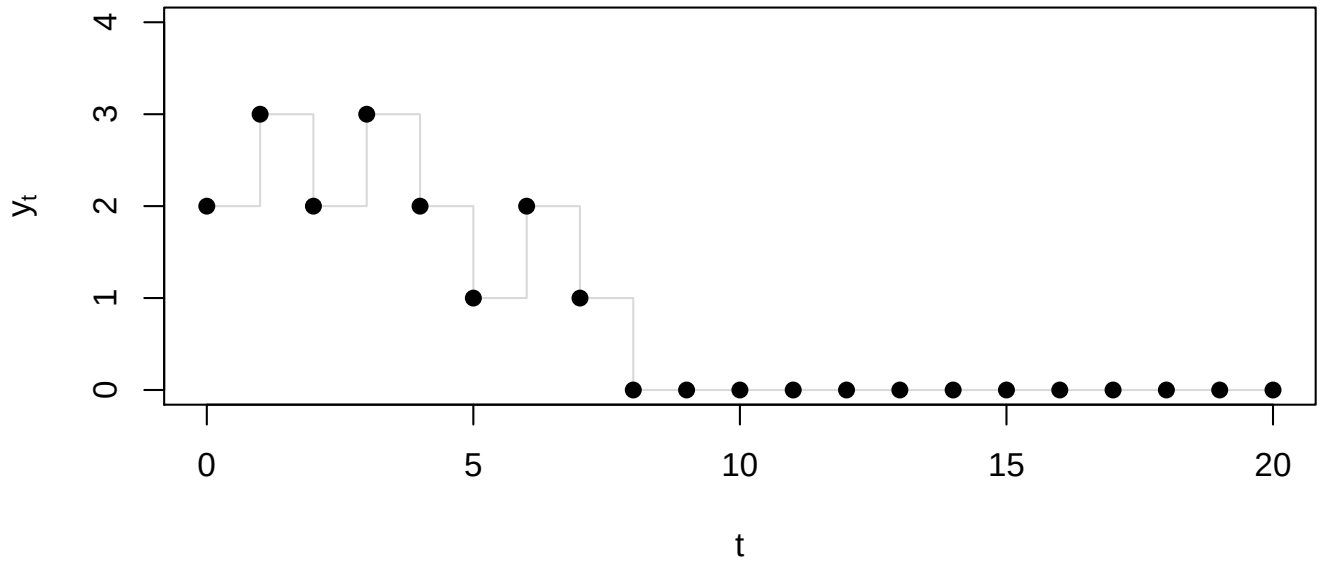
$$P = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

- We perform two simulations of this chain.
- Resulting plots on next slide.

Gamblers example continued

Plots





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Transience and Recurrence

- In the previous Example, the chain would eventually get stuck in state 0 or state 4.
- It follows that after a time, states 1, 2 and 3 will never be revisited.
- Motivated by this, we say that state i is **recurrent** (or **persistent**) if

$$\mathrm{P}(Y_t = i \text{ for some } t \geq 1 | \, Y_0 = i) = 1.$$

- Hence a state is recurrent if and only if eventual return to it is assured.
- If the above probability is strictly less than 1 (i.e. eventual return is not assured) then state i is **transient**.
- From the previous Example, states classified as follows:
  - States 0 and 4 are recurrent.
  - States 1, 2, and 3 are transient.

Communication of States

- Checking to see if a state is recurrent can be done by examining whether there is any possible sequence of transitions that leads to a state from which eventual return is impossible.
- Leads to the concept of **communication** between states.
- State i communicates with j, written $i \to j$ if the chain can ever reach j starting from i.
- Write $p_{ij}^{(t)}$ is the probability of transitioning from state i to state j in a total of t steps.
  - $p_{ij}^{(t)}$ is element i, j of P^t.
- Then $i \to j$ if and only if $p_{ij}^{(t)} > 0$ for some $t \geq 1$.
- States i and j **intercommunicate**, written $i \leftrightarrow j$, if $i \to j$ and $j \to i$.

Irreducibility

- A Markov chain is **irreducible** if all states intercommunicate.
- For such a process, all states will be recurrent.
- E.g. 1: The rainfall Markov chain is irreducible.
- E.g. 2: The gambling Markov chain is not irreducible.
- Irreducibility is a necessary but not sufficient condition for a Markov chain to have stable long-term behaviour.

Periodicity

Introduction

- Consider the simple toy Markov chain with state space $S = \{1, 2\}$ and transition matrix
$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$
- Process simply oscillates between the two states.
- Clearly irreducible since the states intercommunicate...
- ... but the initial conditions are not forgotten (as was the case in the rainfall Example).
- E.g. if chain in state 1 at time $t = 0$, then will also be in state 1 at any even number time point, be it $t = 2$, $t = 4$ or even $t = 100000$.
- This Markov chain is periodic, with period 2.

Periodicity

Formal Definition

- **Period** of state i is greatest common divisor of set $\{t : p_{ii}^{(t)} > 0\}$.
  - $p_{ii}^{(t)}$ is probability that the chain returns to i in t steps (but can also have returned earlier).
- For Markov chain on previous slide:
  - $p_{11}^{(1)} = p_{11}^{(3)} = p_{11}^{(5)} = 0$ and so on for all odd time lags,
  - $p_{11}^{(2)} = p_{11}^{(4)} = p_{11}^{(6)} = 1$ and so forth for even time lags.
- It follows that the set of possible return times for that chain is

$$\{t : p_{ii}^{(t)} > 0\} = \{2, 4, 6, 8, \ldots, \}$$

- That set has greatest common divisor of 2 for $i = 1$ and $i = 2$.
  - Confirms both states have period 2.
- We can show that all states in an irreducible chain have same period.

Aperiodic Markov Chains

- If a state has period 1 then it is called **aperiodic**.
- Note that if $p_{ii} > 0$ then the chain can remain in state i...
- ... so set of possible return times contains $t = 1$...
- so greatest common divisor of 1...
- so such a state is therefore aperiodic.
- If all states in a chain are aperiodic then the chain itself can be referred to as aperiodic.

Ergodic and Regular Markov Chains

- A Markov chain that is irreducible is said to be **ergodic**.
- If it is also aperiodic, then the Markov chain is **regular**.
- Follows that the periodic chain we examined earlier is ergodic but not regular.
- The rainfall Markov chain is regular.
- Regular Markov chains important because guaranteed to have the kind of stable long-term behaviour that we saw in the rainfall examples.

Stationary Distributions

- Recall that $\pi^{(t)}P = \pi^{(t+1)}$.
- In general probability vectors $\pi^{(t)}$ and $\pi^{(t+1)}$ will differ...
- ... but is this bound to be happen?
- If we find a probability vector π^* such that

$$\pi^* P = \pi^*$$

  then setting $\pi^{(t)} = \pi^*$ will give $\pi^{(t+1)} = \pi^*$.
- Can be proved that for any regular Markov chain with finite state space, there must exist a probability vector π^* satisfying above equation.
- π^* called the **stationary distribution** of the Markov chain.
- It is also the **equilibrium distribution** for the chain, since $\pi^{(t)}$ guaranteed converge to π^* as $t \to \infty$ irrespective of initial conditions.

Calculating Stationary Distributions

- Matrix equation for stationary distribution is

$$\pi^* P = \pi^*$$

- This defines a system of simultaneous equations for elements of π^*.
- Additional requirement is that elements of π^* must sum to one.
  - Needed for π^* to be a probability distribution.
- We will provide an example of this being done by hand, and will then move on to computational methods using R.

Example Stationary Distribution for Rainfall Chain

- Recall that Palmerston North summer rainfall Markov chain has transition matrix

$$P = \begin{bmatrix} 0.87 & 0.13 \\ 0.50 & 0.50 \end{bmatrix} .$$

- Stationary distribution $\boldsymbol{\pi}^* = (\pi_1^*, \pi_2^*)$ therefore satisfies

$$\begin{aligned} (\pi_1^*, \pi_2^*) = (\pi_1^*, \pi_2^*)P &= (\pi_1^*, \pi_2^*) \begin{bmatrix} 0.87 & 0.13 \\ 0.50 & 0.50 \end{bmatrix} \\ &= (0.87\pi_1^* + 0.5\pi_2^*,\ 0.13\pi_1^* + 0.5\pi_2^*) \end{aligned}$$

- From this we get the pair of equations

$$\begin{aligned} \pi_1^* &= 0.87\pi_1^* + 0.5\pi_2^* \\ \pi_2^* &= 0.13\pi_1^* + 0.5\pi_2^* \quad . \end{aligned}$$

Example (PN rainfall) continued

- Recall equations are

$$\begin{aligned} \pi_1^* &= 0.87\pi_1^* + 0.5\pi_2^* \\ \pi_2^* &= 0.13\pi_1^* + 0.5\pi_2^* \quad . \end{aligned}$$

- Working with the first equation gives

$$0.13\pi_1^* = 0.5\pi_2^* \Rightarrow 0.26\pi_1^* = \pi_2^*$$

- But this is also what we get from second equation.
- In other words, the two equations are really one in the same.
- Turns out that this is inevitable: $\boldsymbol{\pi}^* P = \boldsymbol{\pi}^*$ will only ever yield one less linearly independent equation than there are states.
- Additional equation provided by

$$\pi_1^* + \pi_2^* = 1 \quad .$$

Example (PN rainfall) continued

- Hence equations to solve are

$$\begin{aligned} 0.26\pi_1^* &= \pi_2^* \\ \pi_1^* + \pi_2^* &= 1 \end{aligned}$$

- Substituting first into second gives

$$\pi_1^* + 0.26\pi_1^* = 1 \Rightarrow (1.26)\pi_1^* = 1$$

- Hence $\pi_1^* = 1/1.26 = 0.794$ (3 d.p.) and $\pi_2^* = 1 - 0.794 = 0.206$.
- Stationary distribution $\boldsymbol{\pi}^* = (0.794,\ 0.206)$ describes probabilities of state membership in the long run.
- Follows that over any extended period, about 79.4% of Palmerston North summer days will be dry.

Matrix Computation of Stationary Distributions

- Finding stationary distributions by hand is tedious if there are more than a few states.
- Therefore useful to look at matrix methods for calculating π^* that can then be implemented in R.
- Texts on linear and matrix algebra tend to be written in terms of column vectors, while our probability vectors are row vectors.
- Therefore convenient to start by transposing equation $\pi^* P = \pi^*$.
- This turns row vectors into column vectors and also reverses the order of multiplication, so we get

$$P^{\mathsf{T}} \pi^{*\mathsf{T}} = \pi^{*\mathsf{T}}$$

where superscript T indicates transposition.

Matrix Computation of Stationary Distributions

Continued

- It follows that

$$P^{\mathsf{T}}\boldsymbol{\pi}^{*\mathsf{T}} = I\boldsymbol{\pi}^{*\mathsf{T}} \Rightarrow (P^{\mathsf{T}} - I)\boldsymbol{\pi}^{*\mathsf{T}} = \mathbf{0}$$

  - I is the $m \times m$ identity matrix (for a state space of size m);
  - $\mathbf{0}$ is vector of zeroes.
- There is linear independence in system of equations, so we can drop the last of them, as generated by the final row of the matrix $(P^{\mathsf{T}} - I)$.
- However, we need to include the additional equation

$$\sum_{i=1}^{m} \pi_i^* = 1.$$

Matrix Computation of Stationary Distributions

Continued

- Putting this all together, we see that π^* solves linear system

$$A{\pi^*}^{\mathsf{T}} = \boldsymbol{b}$$

  - The first $m-1$ rows of A match the first $m-1$ rows of $(P^{\mathsf{T}} - I)$.
  - The last row of A is all ones.
  - $\boldsymbol{b}$ is a vector of zeros except for the last element, which is a one.
- All these ideas illustrated in following Example.

Example: Rainfall Stationary Distribution

Computation using Matrices in R

- Recall

$$P = \begin{bmatrix} 0.87 & 0.13 \\ 0.50 & 0.50 \end{bmatrix}$$

- Hence

$$P^T - I = \begin{bmatrix} 0.87 & 0.50 \\ 0.13 & 0.50 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -0.13 & 0.5 \\ 0.13 & -0.5 \end{bmatrix}.$$

- We replace the final row of this matrix by a row of ones to give

$$A = \begin{bmatrix} -0.13 & 0.5 \\ 1 & 1 \end{bmatrix}.$$

Example (rainfall stationary ditributions) continued

Computation using Matrices in R

- Vector $\boldsymbol{b} = (0, 1)^{\mathsf{T}}$.
- Hence matrix equation to solve is

$$\begin{bmatrix} -0.13 & 0.5 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \pi_1^* \\ \pi_2^* \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

- Multiplying this out gives same simultaneous equations we solved previously.
- We will follow a different path, and find the answer using the `solve` command in R.
  - Function `solve` finds solution to systems of linear equations (and can also be used to find matrix inverses).

Example (rainfall stationary ditributions) continued

R Snippet

```
> P <- matrix(c(0.87,0.5,0.13,0.5),nrow=2)
> diag(2)
     [,1] [,2]
[1,]    1    0
[2,]    0    1
> A <- t(P) - diag(2)
> A
      [,1] [,2]
[1,] -0.13  0.5
[2,]  0.13 -0.5
> A[2,] <- c(1,1)
> A
      [,1] [,2]
[1,] -0.13  0.5
[2,]  1.00  1.0
> b <- c(0,1)
> solve(A,b)
[1] 0.7936508 0.2063492
```

Example (rainfall stationary ditributions) continued

Points to Note

- Command `diag(2)` creates a 2×2 identity matrix.
  - More generally, `diag(m)` will create an $m \times m$ identity matrix.
- Function `solve` can take arguments for the matrix A and the vector $\boldsymbol{b}$ (in that order).
- Naturally the final solution matches that which we found by hand.
- Alternative is to make use of equality of stationary and equilibrium distributions.
- Turns out that every row of the matrix P^t converges to the stationary distribution as $t \to \infty$.
- Calculating P^∞ directly is not possible, but usually P^t with some large value of t will suffice.
- Check if value of t is sufficiently large by comparing the rows of P^t: they should all be the same.

Example (rainfall stationary ditributions) continued

R Snippet

```
> library(expm)
> P%^%5
          [,1]      [,2]
[1,] 0.7950817 0.2049183
[2,] 0.7881473 0.2118527
> P%^%10
          [,1]      [,2]
[1,] 0.7936607 0.2063393
[2,] 0.7936126 0.2063874
> P%^%50
          [,1]      [,2]
[1,] 0.7936508 0.2063492
[2,] 0.7936508 0.2063492
> P%^%1000
          [,1]      [,2]
[1,] 0.7936508 0.2063492
[2,] 0.7936508 0.2063492
```

Example (rainfall stationary ditributions) continued

- In this case the rows of P^{10} provide a fairly good approximation to the stationary distribution.
- P^{50} gives the required probabilities to 8 decimal places.